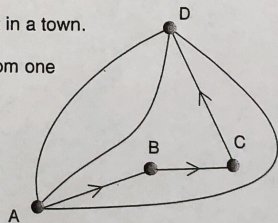


24. Harmonies city music store sales figures for the first four weeks of January 2014 are shown in matrix C and Harmonies suburban stores sales are shown in matrix S.

Harmonies city store				Harmonies suburban store			
	piano	keyboard	organ		piano	keyboard	organ
week 1	2	8	4	week 1	1	4	1
week 2	1	11	2	week 2	1	6	2
week 3	3	9	1	week 3	0	7	1
week 4	2	7	3	week 4	2	9	2

- (a) Find the matrix $C + S$. What does this matrix represent?
- (b) If the city store makes a profit of \$100 on each musical instrument sold, find the matrix $100C$.
- (c) The suburban store makes a profit of \$90 on each instrument sold. Find matrix P if $P = 100C + 90S$.
- (d) What does the matrix P represent?
- (e) Find the total profit made by Harmonies during the month of January.
- (f) If $D = 100C - 90S$ find D and explain the information contained in matrix D.
- (g) What does the "-" indicate in the entry d_{42} .

28. The following map indicates road links joining locations of interest in a town.



- (a) Construct matrix P showing the number of direct road links from one location of interest to another.

- (b) State the total number of different direct journeys that can be taken between the locations of interest in the town.

- (c) Determine P^2 .

- (d) Matrix P is classified as a one-stage route matrix. Classify matrix P^2 and explain what information it holds.

- (e) What does the entry located at p_{31}^2 tell you? (f) Determine P^3 . What does this matrix represent?